



Disaster Recovery Process Document

Customer:

Site Name:

Date:

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1. Project Information

Site Name	
Address	
Owner	
Representative	
Contractor	
Sub-Contractor	
Date	

2. Introduction

This document provides a basic disaster recovery plan. Each customer, project, and site is unique, so specific procedures may vary. It outlines steps to ensure continuity and recovery of a cluster SCADA system, including OPC services, telemetry radio connections, and local SQL datalogging. The plan also incorporates key stakeholders, critical spares, and pre-identified system risks

2.1 REQUIREMENTS

- a) Access to the site/server room
- b) Backup power in the server room
- c) Cold standby server with latest licenced software loaded.
- d) USB Drive with the latest backups
- e) Admin User Login Credentials
- f) Network Diagram and IP addresses
- g) Maintain a list of key stakeholders for disaster response
- h) Ensure availability of critical spares (PSU, hard drives, network cards, graphic cards, RAM, OS keys, passwords, installation disks, switches)
- i) Document site-specific environmental controls (temperature regulation, humidity)

2.2 Stakeholders

Maintain a list of personnel responsible for disaster response:

- a. SCADA Administrator / Lead Engineer – Responsible for recovery of SCADA servers and software
- b. IT / Network Administrator – Ensures connectivity, firewall rules, switches, and remote access
- c. Database Administrator – Responsible for SQL server and data integrity
- d. Operations Manager / Process Owner – Confirms system operation and verifies critical process continuity
- e. Maintenance Engineer / Onsite Support – Handles hardware swaps, spare parts, and server power issues
- f. Remote Support Contact (Adroit Technologies / OEM) – Escalation for technical support

2.2 COLD STANDBY SERVER ACTIVATION

In the event of a disaster, follow the steps below to activate the cold standby server:

- a) Keep the standby-server configuration, software, database, and project configuration up to date and test regularly.
- b) Keep some basic server spares onsite.
- c) Use soft licence keys instead of HASP USB keys.
- d) Power on the cold standby server and ensure it is connected to the network.

- e) Configure the necessary network settings to match the production server.
- f) Install and configure any required software or services on the cold standby server.
- g) Restore the latest project backup from the removable USB drive to the cold standby server.
- h) Verify the connectivity between the cold standby server and the other system components.

Ensure all spares are readily available for server replacement or component failure

2.3 PROJECT BACKUP AND RECOVERY

Ensure the following steps are taken to manage project backups:

- a) Regularly back up the project files to the removable USB drive (Scheduled automated backups).
- b) Store the removable USB drive in a secure location onsite.
- c) Connect the removable USB drive to a UPS backup power source and ensure it is accessible to the switch.
- d) Test the restoration process periodically to ensure the backup files are valid and can be successfully recovered.
- e) Maintain multiple backup locations if possible (offsite or cloud) for added resilience
- f) Include incremental backups for minimal data loss

2.4 DATA LOG CONNECTOR AND DATA WAREHOUSE SERVER

Verify the health and connectivity of the data log connectors and data warehouse server as follows:

- a) Periodically check the data log connectors to ensure they are functioning correctly.
- b) Validate that the data log connectors are successfully sending data to the data warehouse server.
- c) Regularly test the data warehouse server's ability to receive and process data from the log connectors.
- d) Monitor data replication and storage to confirm the integrity of the data.

2.5 USER SECURITY AND LOG ON PROCEDURES

Maintain user security and log on functionality by adhering to the following measures:

- a) Regularly review and update user accounts, access levels, and privileges.
- b) Ensure that user credentials are securely stored and encrypted.
- c) Test the log on procedure periodically to verify its functionality.
- d) Establish a process for password recovery and account lockout in case of security breaches.
- e) Regular review of users, privileges, and credentials
- f) Establish password recovery and lockout procedures

2.6 REMOTE UI SERVER AND COLD STANDBY SERVER ACCESS

Enable the remote UI server to access the cold standby server with the following steps:

- a) Configure network access rules to allow the remote UI server to connect to the cold standby server.
- b) Test the connectivity between the remote UI server and the cold standby server.
- c) Ensure that all necessary software and services are installed and properly configured on the cold standby server to support remote access.
- d) Network rules for remote access
- e) Test connectivity
- f) Ensure proper software installed for remote access

2.7 SYSTEM HEALTH CHECKS

Regularly perform system health checks to ensure the SCADA system and devices are functioning optimally:

- a) Develop a monitoring and alerting system to identify any abnormalities or failures in the system.
- b) Schedule regular system health checks, including device polling and status verification.
- c) Keep all devices and software up to date with the latest patches and updates.
- d) Perform periodic tests to confirm the responsiveness and accuracy of device communication.
- e) Monitoring and alerting
- f) Device polling and status verification
- g) Apply software patches and firmware updates

2.8 SYSTEM SPARES

Maintain the following critical spare components onsite to reduce downtime:

- PSU (Power Supply Units)
- Hard Drives (OS, SQL, SCADA backups)
- Network Cards & Switches
- Graphic Cards
- RAM Modules
- Cold standby servers (if multiple servers are available)
- Operating System and SCADA software keys
- Installation media/disks

3. Conclusion

This disaster recovery document provides guidelines to ensure the continuity and recovery of the cluster SCADA system in case of a disaster. Regular testing, monitoring, and backup management will contribute to a reliable and resilient SCADA system.

4. Approval

Signed at _____ on this date _____

Name & Surname

Project Engineer

Signed at _____ on this date _____

Name & Surname

Project Sponsor